PFP II Landau theoretical minimum: worked instructor key

Worked instructor key: PFP II

AuthorityEffect: None. Attempt the quiz first. The key explains the expected reasoning and a 100-point rubric. It is an educational companion to the scoped proofs and implementation, not an additional publication claim.

Q1. Declaration (6 points)

The carrier is the ordered finite vertices (a, b, c) with the three indexed undirected edges specified in the quiz. Coefficients are $\mathbb{Z}_2$; allowed moves assign one sign to each vertex, equivalently one bit α_v. Acceptance requires $\alpha_u \oplus \alpha_v = c_e$ on every edge. Bit zero requires equal bits and bit one opposite bits.

A name or drawing supplies no face, orientation-dependent rational report, or physical Hamiltonian. A bracket encoding needs an interpretation before it denotes any of these objects.

Credit: carrier and edge identity 2; coefficients and moves 2; acceptance equation and no undeclared structure 2.

Q2. All gauges (6 points)

With $\alpha_a = 0$, the first edge forces $\alpha_b = 1$, and the second forces $\alpha_c = 0$. The closing edge checks $0 \oplus 0 = 0$. The three equations are $0 \oplus 1 = 1$, $1 \oplus 0 = 1$, and $0 \oplus 0 = 0$.

Without root-fixing, the gauges are 010 and 101. XORing two valid assignments gives a bit constant on each connected component, so these are the only two on this connected graph. Each additional consistent connected component supplies an independent global flip. In general a consistent graph with k components has 2^k gauges, including the empty graph's one empty assignment.

Credit: rooted calculation and all edge checks 3; two gauges and component freedom 3.

Q3. Indexed obstructions (6 points)

For the modified triangle, use `((0,a,b),(1,b,c),(2,c,a))`. It is continuous and closed, and its parity is $1 \oplus 1 \oplus 1 = 1$. XORing the three endpoint equations would give zero on the left and one on the right.

For one vertex a, a parity-one edge aa has witness `((0,a,a),)` and asks $\alpha_a \oplus \alpha_a = 1$. For two vertices, let edge 0 join a, b with parity zero and edge 1 join them with parity one. The witness `((0,a,b),(1,b,a))` has parity one. Distinct edge IDs are essential: the same undirected edge traversed out and back contributes its bit twice and cannot be this obstruction.

Removing loops excludes a valid declared input and changes the grammar.

Credit: triangle witness and contradiction 2; loop 2; parallel-edge witness and grammar distinction 2.

Q4. Rational near miss (8 points)

The binary parity of $(1, 1, 0)$ is zero. Its rational cycle sum is two. An exact potential would require

$$(p_b - p_a) + (p_c - p_b) + (p_a - p_c) = 0,$$

so these rational reports are not exact.

If the changes are $\delta_0, \delta_1, \delta_2$, their sum must be -2. The triangle inequality gives $\sum_i |\delta_i| \geq |\sum_i \delta_i| = 2$. Change the closing report from zero to -2, leaving reports $(1, 1, -2)$. The potential $(p_a, p_b, p_c) = (0, 1, 2)$ attains this repair with cost two. No two-dimensional filling was needed.

Credit: parity versus rational sum 2; telescoping obstruction 2; lower bound 2; attaining reports and potential 2.

Q5. Balance theorem and construction (10 points)

Necessity. XOR the endpoint equations around any closed walk, counting occurrences with multiplicity. Every arrival matches a departure around the closed walk, so the vertex terms cancel, including for a loop or a repeated edge. The edge XOR must be zero.

Sufficiency. Choose a root in each component and set its bit to zero. For each other vertex, define its bit as the parity of a root-to-vertex path. Any two such paths give a closed walk by reversing the second, so their parities agree. Append an edge uv to a root-to-u path; path independence proves $\alpha_v = \alpha_u \oplus c_{uv}$. This proves every edge equation. Isolated vertices are free; an odd loop or disagreeing parallel edges contradict the premise directly.

Breadth-first construction stores adjacency, assigned bits, and tree parents. Process each vertex and edge a bounded number of times. The first conflicting edge

plus the tree paths supplies a closed odd walk; if none conflicts, return the assignment. Producing that one walk costs at most linear extra work. Independent verifiers check the returned witness. The resulting $O(|V|+|E|)$ algorithm is the baseline; enumerating $2^{|V|}$ assignments exercises receipts rather than improving this classification algorithm.

Credit: necessity 3; sufficiency with path independence and components 4; constructive witnesses and linear versus exponential baseline 3.

Q6. Matrix (8 points)

The graph bits for ab, bc, ca are $1, 1, 0$, because the first two matrix entries are positive and the third negative. Entrywise multiplication gives

$$DHD = \begin{pmatrix} 0 & -1 & -1 \\ -1 & 0 & -1 \\ -1 & -1 & 0 \end{pmatrix}.$$

Multiplication by signs leaves zero unchanged, so a zero off-diagonal entry imposes no constraint. The exact adapter refuses floats, nonsquare arrays, and nonsymmetric arrays with an input-contract error; this is not an odd-walk proof for a well-formed instance.

An all-positive triangle excludes diagonal vertex signs in the fixed basis. It does not exclude an unrestricted global eigenbasis: every finite Hermitian matrix has a unitary diagonalization. Finding and using that basis can be computationally costly.

Credit: bits and computed matrix 4; zero and input refusals 2; transformation boundary 2.

Q7. Exponential (10 points)

Choose $a \geq \max_i H'_{ii}$. Then $B = aI - H'$ is entrywise nonnegative: its off-diagonal entries are $-H'_{ij} \geq 0$ and its diagonal entries are $a - H'_{ii} \geq 0$. Products of entrywise nonnegative matrices remain entrywise nonnegative. Because aI commutes with B,

$$e^{-\beta H'} = e^{-\beta a} e^{\beta B} = e^{-\beta a} \sum_{n=0}^{\infty} \frac{\beta^n B^n}{n!}.$$

For $\beta \geq 0$, all coefficients and the scalar factor are nonnegative. In any submultiplicative matrix norm the terms are bounded by $(\beta\|B\|)^n/n!$, whose sum converges. Thus the entrywise limit is nonnegative. At $\beta = 0$ it is the identity, so the statement is nonnegativity, not strict positivity.

An efficiency claim needs a specified sampling procedure and its cost, including relevant mixing or convergence behavior. Observable estimates also require their

preserved numerators and variance or error analysis. Matrix-entry positivity alone supplies none of these guarantees.

Credit: correct shift and matrix powers 3; series identity and coefficients 3; convergence and beta-zero case 2; computational boundary 2.

Q8. Two checks, then one (8 points)

Call	Recorded candidate checks	Status	Witness	Remaining interval
Initial budget 2	`0:000 false; 1:001 false`	`UNKNOWN`	None	$[2, 8)$
Resume budget 1	`2:010 true`	`GAUGE_VERIFIED`	Gauge 010	$[3, 8)$

Candidate 000 fails the first edge. Candidate 001 also fails it; it additionally disagrees with the closing parity. Candidate 010 satisfies every edge equation. The resumed call starts at two and ends at cursor three.

Verification of the second receipt requires the first UNKNOWN receipt, bound to the same graph and key. Its earlier checks are replayed, and the chain must begin at zero and reach the second start index without gaps. The second segment is then checked, including its witness and digest. A correct digest alone does not establish prior coverage.

Credit: initial trace 3; resumed trace 3; full prior-chain obligation 2.

Q9. Continuation attacks (6 points)

Starting at two without prior receipts is refused: the verified prefix cursor is zero. A caller-supplied count or frontier cannot supply coverage. Flipping a verdict and rehashing is rejected when replay disagrees with that verdict, even though the new digest matches the forged record.

Changing the basis changes the full presentation digest, so the old receipt cannot certify continuation under the new key. Identical Hamiltonian bytes are insufficient. The implementation also binds graph bytes and checks checker version, transformation grammar, segment order, candidate bits, verdicts, and frontier bounds.

A valid contiguous UNKNOWN chain beginning at zero supplies the prior coverage. Retrieval compares the complete declared key but returns a proposal; recheck its bits against the current graph. Neither successful retrieval nor gauge verification proves that the declared physical adapter preserves the Hamiltonian, measure, or observables.

Credit: missing prefix 2; rehashed false verdict 2; changed context and retrieval/adapter boundary 2.

Q10. Full batch and zero budget (6 points)

From zero with budget eight, the implementation completes all eight checks in its first batch. The valid candidates are indices two (010) and five (101); it selects the first in canonical order, 010. The status is `GAUGE_VERIFIED`, cursor eight, and suffix $[8, 8)$, an empty interval. Finding a witness at index two does not make the cursor three for this call.

Workers one and four produce the same ordered checks, chosen witness, and receipt. Scheduling does not enter the receipt's mathematical content. Q8's smaller budgets limit its first batch to two checks and its next to one, so its successful cursor is three.

An initial zero-budget call on this nonempty candidate space records no checks, has no witness, returns UNKNOWN, and leaves $[0, 8)$. It establishes no negative result and supplies no advancing prefix for continuation.

Credit: witness, checks, and cursor 3; worker independence and Q8 contrast 2; zero-budget limit 1.

Q11. Pairing (4 points)

On a paired subset P, the map must take values in P and satisfy $I(I(x)) = x$, $I(x) \neq x$, opposite signs, and equal magnitudes. These give disjoint canceling two-element orbits; any unpaired survivors must be assessed separately.

For $+2, -1, -1$, the total is zero but each negative magnitude is one and the only positive magnitude is two. No required partner exists. Regrouping both negative terms into a single contribution changes the carrier and permitted operation.

For real finite beta and Hermitian H on a nonzero finite-dimensional space, $\operatorname{Tr} e^{-\beta H} = \sum_j e^{-\beta \lambda_j} > 0$. Pairing every term of an absolutely convergent expansion would instead make its sum zero. Absolute convergence is needed to regroup infinitely many terms this way.

Credit: map obligations 1; counterexample and regrouping boundary 2; positive trace with convergence condition 1.

Q12. Target path (8 points)

The modes are $(\uparrow 0, \uparrow 1, \uparrow 2, \downarrow 0, \downarrow 1, \downarrow 2)$. Using zero-based mode indices:

Hop	Changed modes	Occupied modes strictly between	Sign
`110;111 -> 101;111`	1, 2	0	+1

Hop	Changed modes	Occupied modes strictly between	Sign
`101;111 -> 011;111`	0, 1	0	$+1$
`011;111 -> 110;111`	0, 2	1	-1

Each state has two doubly occupied sites, hence energy $2 \cdot 2 = 4$. Including the final root again, the dwell times sum to β; the integrand is the constant $e^{-4\beta} = e^{-1}$. The ordered three-jump simplex has volume $(1/4)^3/3! = 1/384$. The sign product is negative, so the weight is $-e^{-1}/384$.

There are 64 occupation states, but arbitrarily long closed hopping sequences are possible. Repeating a nonzero two-hop out-and-back loop already supplies unbounded lengths. Integrating times does not identify these different state sequences.

Credit: hopping signs 3; energies and dwell-time integrand 2; simplex and signed weight 2; finite states versus unbounded paths 1.

Q13. All lengths (8 points)

For an n-hop sequence, A is its positive time integral; each nonzero hopping magnitude is one.

Length	Bound or exclusion	Reason
0	$A \geq e^{-3/2}$	$E \leq 6$, beta $1/4$
1	No closed sequence	No diagonal hopping
2	$A \geq e^{-3/2}/32$	Global energy bound and $\beta^2/2! = 1/32$
3, positive	$A \geq e^{-1/2}/384$	Enumerated positive paths have $E \leq 2$
$n \geq 4$	$A \leq \beta^n/n! \leq 1/6144$	$E \geq 0$; ratio $\beta/(n+1) < 1$

The target therefore satisfies

$$\frac{1}{6144} < \frac{e^{-1}}{384} < \frac{e^{-1/2}}{384}, \qquad \frac{e^{-1}}{384} < \frac{e^{-3/2}}{32} < e^{-3/2}.$$

The first lower comparison is $e^{-1} > 1/16$; the short-path comparisons follow from the stated strict rational enclosures, rather than rounded decimal evaluations.

For $x \in [0, 1]$, successive magnitudes in $\sum_{k\geq 0}(-x)^k/k!$ decrease and tend to zero. Odd truncations bound e^{-x} below; even truncations bound it above. The checker encloses e^{-1}, $e^{-1/2}$, and $e^{-3/4}$. Its positive lower bound for the last can

be squared to bound $e^{-3/2}$. Sufficiently tight rational bounds establish the strict comparisons by fraction arithmetic.

This covers every length and excludes an individual positive path of equal magnitude. It combines finite enumeration and analytic bounds. The bound 1/6144 concerns one sequence; multiplying by a sequence count or proving convergence is required for a total-tail estimate. Time-labelled maps, collective regroupings, other bases, and other estimators require different arguments.

Credit: all five length classes and reasons 4; rational strict comparisons and decreasing long bound 2; individual-path and representation boundaries 2.

Q14. Audit (6 points)

One acceptable exact claim is: at the specified mode order and parameters, the designated negative integrated rooted path has no positive individual path of equal magnitude. Evidence is the Section 6 bounds plus `code/exact_fock_gap.py` and `code/tests/test_exact_fock_gap.py::test_exact_gap_covers_every_len`. The execution receipt records what actually ran; neither test passage nor the path obstruction classifies every physical representation.

Proposed conclusion	Correct assessment
Two failures prove no gauge	UNKNOWN with the untested suffix; complete coverage or a directly verified odd walk is missing
The gap excludes every sign-free representation	It excludes the stated individual integrated-path pairing, under the fixed parameters
The manifest verifies mathematics	It checks hashes and file census; proofs and checkers address mathematical claims
Fallback verifies the native executable	It checks the Python route; native discovery may skip until a binary is built and exercised

Credit: exact claim, domain, and named evidence 2; one point per corrected conclusion 4. Retain the scoped result rather than demanding evidence for a larger, unclaimed physical classification.